

Relocate

Full System Documentation

Presentation & Technical Reference

Moving & relocation platform — Mombasa & Nairobi, Kenya

M-Pesa payments · Driver portal · Live GPS · Admin dashboard

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github.com/techwizh/relocationweb

Executive summary

Relocate is a web platform for booking moving vehicles in Kenya.

Customers book online, pay via M-Pesa, chat with drivers, and track moves live.

Drivers register, get approved by admin, receive jobs, and share GPS.

Admins manage the landing page, approve drivers, assign bookings, and monitor fleet.

Live stack: Vercel (frontend) + Render (backend API) + Neon (PostgreSQL)

Website: Vercel URL | **API:** relocationweb-api.onrender.com

Relocate — Full System Documentation

Book moving vehicles in **Mombasa** and **Nairobi**, Kenya. Own-fleet model with M-Pesa payments, in-app chat, live GPS tracking, driver portal, and admin dashboard.

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Live URLs

Service	URL	Purpose
Website (public)	Your Vercel URL (e.g. <code>https://relocationweb.vercel.app</code>)	Customers, drivers, admin UI
Backend API	<code>https://relocationweb-api.onrender.com</code>	API, database, M-Pesa, uploads
Health check	<code>https://relocationweb-api.onrender.com/api/health</code>	Verify backend is running
Source code	<code>https://github.com/techwizh/relocationweb</code>	Git repository (not the live site)

Important: GitHub shows the README and code. The live website is always the **Vercel URL**.

Architecture

Production uses a **split deploy**:



Local development runs everything on one machine (`localhost:3000`) with no `API_URL` set.

Tech stack

Layer	Technology
Framework	Next.js 16 (App Router)
Language	TypeScript
Styling	Tailwind CSS 4
Database	PostgreSQL via Prisma 6
Maps	Leaflet / react-leaflet
Payments	Safaricom Daraja M-Pesa (STK Push)
Frontend host	Vercel
Backend host	Render (free tier)
Database host	Neon (free tier)
Image storage	<code>public/uploads/</code> locally; Vercel Blob in production (optional)

User roles & flows

Customer

- Visit landing page → **Book a move**
- Choose city, sub-county, ward, vehicle, date, contact details
- Pay with M-Pesa (or skip if `SKIP_MPESA_PAYMENT=true`)
- Chat with driver, track live GPS
- Optional: register at `/register` → **My account** lists all bookings

Driver

- Register at `/driver/register` (photos, license, vehicle)
- Wait for admin approval
- Log in at `/login` (Driver tab)
- **Driver portal**: toggle availability, accept assigned jobs, update status, share GPS, chat

Admin

- Log in at `/admin/login`
 - **Dashboard** — stats overview
 - **Content** — edit landing page text and upload hero/gallery images
 - **Drivers** — approve/reject applications, set internal contact phone
 - **Bookings** — assign paid bookings to approved drivers
 - **Fleet** — live map of driver GPS
-

Pages & routes

Public

Path	Description
/	Landing page (CMS-driven)
/book	New booking form
/book/[id]/pay	M-Pesa payment
/book/[id]/chat	Booking chat
/book/[id]/track	Live driver map
/book/access-denied	Shown when booking access fails
/login	Customer or driver login
/register	Customer registration
/driver/register	Driver application
/driver/register/success	Post-registration confirmation

Customer (requires login)

Path	Description
/account	My bookings, chat/track links

Driver (requires login + approved profile)

Path	Description
/driver/dashboard	Jobs, availability, GPS sharing

Admin (requires admin session)

Path	Description
/admin	Dashboard stats
/admin/content	Landing page CMS
/admin/drivers	Review driver applications
/admin/bookings	Assign drivers to bookings
/admin/fleet	Live fleet map
/admin/login	Admin login

API reference

All API routes live on **Render** in production. Vercel proxies `/api/*` to Render.

Health

Method	Path	Auth	Description
GET	<code>/api/health</code>	None	Service health check

Landing & CMS

Method	Path	Auth	Description
GET	<code>/api/landing</code>	None	Public landing page content
GET	<code>/api/admin/content</code>	Admin	Get CMS content
PUT	<code>/api/admin/content</code>	Admin	Save CMS content
POST	<code>/api/admin/upload</code>	Admin	Upload image

Bookings

Method	Path	Auth	Description
POST	<code>/api/bookings</code>	Optional customer	Create booking
GET	<code>/api/bookings/[id]</code>	Guest/customer cookie	Booking details for pay page
GET	<code>/api/bookings/[id]/access</code>	Booking access	Access check for chat/track
POST	<code>/api/bookings/[id]/pay</code>	Booking access	Initiate M-Pesa STK Push
GET	<code>/api/bookings/[id]/payment-status</code>	Booking access	Poll payment status
GET	<code>/api/bookings/[id]/messages</code>	Booking access	List chat messages
POST	<code>/api/bookings/[id]/messages</code>	Booking access	Send chat message
GET	<code>/api/bookings/[id]/location</code>	Booking access	Latest driver GPS for booking

M-Pesa

Method	Path	Auth	Description
POST	<code>/api/mpesa/callback</code>	Safaricom	Payment webhook (Render URL only)

Customer auth

Method	Path	Auth	Description
POST	/api/customer/register	None	Register
POST	/api/customer/login	None	Login
POST	/api/customer/logout	Customer	Logout
GET	/api/customer/me	Customer	Current user
GET	/api/customer/bookings	Customer	My bookings list

Driver auth & portal

Method	Path	Auth	Description
POST	/api/drivers/register	None	Driver application
POST	/api/driver/login	None	Login
POST	/api/driver/logout	Driver	Logout
GET	/api/driver/me	Driver	Current driver user
GET	/api/driver/dashboard	Driver	Jobs + profile for portal
POST	/api/driver/availability	Driver	Toggle available/offline
POST	/api/driver/location	Driver	Submit GPS coordinates
POST	/api/driver/jobs/[bookingId]/status	Driver	Advance job status

Admin

Method	Path	Auth	Description
POST	/api/admin/login	None	Admin login
POST	/api/admin/logout	Admin	Logout
GET	/api/admin/stats	Admin	Dashboard counts
GET	/api/admin/bookings	Admin	Bookings + approved drivers
POST	/api/admin/bookings/[id]/assign	Admin	Assign driver to booking
GET	/api/admin/drivers	Admin	Driver list (?status=PENDING)
POST	/api/admin/drivers/[id]/review	Admin	Approve/reject driver
PATCH	/api/admin/drivers/[id]/contact	Admin	Set driver internal phone
GET	/api/admin/fleet	Admin	Live driver locations

Database schema

PostgreSQL managed by Prisma. Key models:

Model	Purpose
User	Customers, drivers, admins (role enum)
DriverProfile	Driver status, availability, photos
Vehicle	Driver vehicle details
Booking	Move details, status, pricing
Payment	M-Pesa payment records
ChatMessage	Per-booking chat
LocationUpdate	Driver GPS history
LandingPageContent	CMS JSON for home page

Booking status flow



Driver advances: **Assigned** → **En route** → **Loading** → **In transit** → **Delivered**

Vehicle types & base prices (KES)

Type	Base price
Van	3,500
Pickup	5,500
Canter (3-ton)	9,000
Lorry (7-ton+)	15,000

Cities

- **Mombasa** — Mvita, Nyali, Kisauni, Likoni, Changamwe, Jomvu
- **Nairobi** — Westlands, Dagoretti, Lang'ata, Kibra, etc.

Authentication & sessions

All sessions use **httpOnly cookies** signed with `ADMIN_SESSION_SECRET` .

Cookie	Role	Max age
<code>relocate_admin_session</code>	Admin	8 hours
<code>relocate_customer_session</code>	Customer	30 days
<code>relocate_driver_session</code>	Driver	7 days
<code>relocate_booking_{id}</code>	Guest booking access	90 days

Split deploy rule: `ADMIN_USERNAME` , `ADMIN_PASSWORD` , and `ADMIN_SESSION_SECRET` must be **identical on Render and Vercel** so login cookies work across the proxy.

Guest booking cookies are set when a booking is created, allowing chat/track/pay without an account.

M-Pesa payments

- Customer submits booking → status `PENDING_PAYMENT`
- `/book/[id]/pay` triggers STK Push via Daraja API
- Customer enters PIN on phone
- Safaricom calls `MPESA_CALLBACK_URL` on **Render**
- Booking status → `PAID` ; customer redirected to chat

Sandbox testing

Setting	Value
<code>MPESA_ENV</code>	<code>sandbox</code>
Test phone	<code>254708374149</code>
Test PIN	<code>174379</code>
Shortcode	<code>174379</code>

Set `SKIP_MPESA_PAYMENT=true` to bypass payment during testing (booking goes straight to `PAID`).

Production (live money)

- Complete Safaricom **Go Live** on [Daraja portal](#)
 - Set `MPESA_ENV=production` on **Render**
 - Use live shortcode, passkey, consumer key/secret
 - Set `MPESA_CALLBACK_URL=https://relocationweb-api.onrender.com/api/mpesa/callback`
 - Set `SKIP_MPESA_PAYMENT=false`
-

File uploads

Environment	Storage
Local dev	<code>public/uploads/</code>
Render (with <code>BLOB_READ_WRITE_TOKEN</code>)	Vercel Blob (public URLs)
Render (without token)	<code>public/uploads/</code> (ephemeral on Render — use Blob for production)

Allowed: JPG, PNG, WebP, GIF — max 5 MB each.

Used for: driver profile/license photos, vehicle photos, admin hero/gallery images.

Environment variables

Render (backend) — required

Variable	Description
<code>DATABASE_URL</code>	Neon PostgreSQL connection string
<code>ADMIN_USERNAME</code>	Admin login username
<code>ADMIN_PASSWORD</code>	Admin login password
<code>ADMIN_SESSION_SECRET</code>	Cookie signing secret (32+ chars)
<code>SKIP_MPESA_PAYMENT</code>	<code>true</code> to skip M-Pesa in testing
<code>MPESA_ENV</code>	<code>sandbox</code> or <code>production</code>
<code>MPESA_CONSUMER_KEY</code>	Daraja consumer key
<code>MPESA_CONSUMER_SECRET</code>	Daraja consumer secret
<code>MPESA_SHORTCODE</code>	Paybill/till (174379 sandbox)
<code>MPESA_PASSKEY</code>	Daraja passkey
<code>MPESA_CALLBACK_URL</code>	Render callback URL
<code>MPESA_ACCOUNT_REFERENCE</code>	<code>Relocate</code>
<code>MPESA_TRANSACTION_DESC</code>	<code>Move booking</code>

Vercel (frontend) — required

Variable	Description
<code>API_URL</code>	Render backend URL (no trailing slash)
<code>ADMIN_USERNAME</code>	Must match Render
<code>ADMIN_PASSWORD</code>	Must match Render
<code>ADMIN_SESSION_SECRET</code>	Must match Render
<code>SKIP_MPESA_PAYMENT</code>	<code>true</code> during testing

Do not set `DATABASE_URL` on Vercel.

Optional

Variable	Where	Description
BLOB_READ_WRITE_TOKEN	Render	Vercel Blob for persistent uploads
NEXT_PUBLIC_MAPBOX_TOKEN	Either	Map provider (future use)

Deployment

See also: [SPLIT-DEPLOY.md](#)

Render (backend)

```
Build: npm install && npx prisma generate && npx prisma migrate deploy && npm run build
Start: npm start
Health: /api/health
```

Vercel (frontend)

```
Build: prisma generate && next build
Env: API_URL + admin secrets (no DATABASE URL)
```

Vercel `next.config.ts` rewrites `/api/:path*` → `${API_URL}/api/:path*`.

Custom domain

- Add domain in **Vercel** → Settings → Domains (website)
 - Keep M-Pesa callback on **Render** URL (or proxy if using same domain)
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Local development

```
cd C:\Users\ADMIN\Projects\relocate
npm install
```

Create `.env` from `.env.example` . For local DB use Neon URL or PostgreSQL:

```
DATABASE_URL="postgresql://..."
# Do not set API_URL locally
```

```
npm run db:deploy
npm run dev
```

Open <http://localhost:3000>

Useful scripts

Command	Description
<code>npm run dev</code>	Start dev server
<code>npm run build</code>	Production build
<code>npm run db:migrate</code>	Create migration (dev)
<code>npm run db:deploy</code>	Apply migrations (prod/local)
<code>npm run db:studio</code>	Prisma Studio GUI

Project structure

```
|
├── relocate/
├── └── prisma/
│   ├── schema.prisma      # Database models
│   └── migrations/        # PostgreSQL migrations
├── public/uploads/        # Local image uploads
├── └── src/
│   ├── app/
│   │   ├── api/           # Backend API routes
│   │   ├── admin/         # Admin dashboard pages
│   │   ├── account/       # Customer account
│   │   ├── book/          # Booking flow
│   │   ├── driver/        # Driver portal
│   │   ├── login/ register/ # Auth pages
│   │   └── page.tsx        # Landing page
│   ├── components/        # React UI components
│   └── lib/
│       ├── admin-auth.ts   # Admin session
│       ├── customer-auth.ts # Customer session
│       ├── driver-auth.ts  # Driver session
│       ├── booking-access.ts # Guest booking cookies
│       ├── mpesa.ts        # Daraja API
│       ├── prisma.ts       # Database client
│       ├── uploads.ts      # Image upload handler
│       ├── api-server.ts   # Server-side API fetch (split deploy)
│       └── api-url.ts      # API_URL helper
├── render.yaml            # Render deploy blueprint
├── vercel.json            # Vercel build config
├── SPLIT-DEPLOY.md        # Deploy walkthrough
└── SYSTEM.md              # This document
```

Troubleshooting

Problem	Likely cause	Fix
GitHub shows README only	GitHub is code storage	Use Vercel URL for the website
<code>/api/health</code> fails on Render	Build failed or service sleeping	Check Render logs; wait 30–60s on free tier
Admin login fails on Vercel	Secrets don't match Render	Sync <code>ADMIN_PASSWORD</code> and <code>ADMIN_SESSION_SECRET</code>
Failed to parse URL from <code>/api/...</code>	Server fetch without origin	Fixed in latest code; redeploy Render
M-Pesa STK push fails	Wrong keys or callback URL	Check Daraja credentials; callback must be Render HTTPS URL
Image upload fails on Render	No persistent disk	Add <code>BLOB_READ_WRITE_TOKEN</code> on Render
First request very slow	Render/Neon free tier cold start	Normal; upgrade plan for always-on
Booking chat "forbidden"	No guest cookie or not paid	Complete payment first; cookies must match domain
Build fails on Vercel with Prisma	<code>DATABASE_URL</code> set on Vercel	Remove <code>DATABASE_URL</code> from Vercel env

Security checklist (before public launch)

- ☐ Change `ADMIN_PASSWORD` from default/example
 - ☐ Use a long random `ADMIN_SESSION_SECRET`
 - ☐ Set real M-Pesa credentials and `SKIP_MPESA_PAYMENT=false`
 - ☐ Do not commit `.env` to GitHub
 - ☐ Review approved drivers before they receive jobs
 - ☐ Consider custom domain + HTTPS on Vercel
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